

Facial recognition to catch criminals

Voice of Big Data (VOBD), which is a US-based data analytics and business intelligence solutions firm with offshore office in India, said it would soon launch 'Facial Signature.'

VOBD would perhaps be the first one in India to launch 3D facial detection product that could aid police and investigating officials to recognize criminals by using the data analytics tools. The company has helped a few police stations in India to solve their cases within a few hours," according to its chief executive Srinu Kishan.

The company stated that Facial Signature is nearly 90 per cent accurate and the algorithms involved in this application can identify the 80 nodal points on a person's face. It will also be possible to identify a person in a disguise or with a *burkha*.

Robot to remove blood clots from brain

Blood clots can form within the brain whenever a blood vessel ruptures. The blood seeps out of the vessel and forms a clot. In most cases, brain blood clots, or intracerebral hemorrhage, can prove to be fatal for the patient.



Blood clot being removed by the robot created by researchers of Vanderbilt University

As per crazyengineers.com, a robot getting ready could bring a change to this situation by helping the doctors in sucking the blood out of the brain where the clot is formed. A robot that can help doctors in saving lives is being developed by research engineers at the Nashville's Vanderbilt University.

A tube is inserted into the brain with the aid of a CT scan. The robot then inserts a very thin tube within this tube to suck the blood from the clot. The experiments have revealed that almost 92 per cent of the blood clots can be removed using the new robot. The researchers are now working on adding an ultrasound imaging technique to the robot to form a computerised model of the clot.

3D-printed liquid metal may lead to bendy electronics

Researchers at North Carolina State University used a mixture of two metals – gallium and indium – that are liquid at room temperature but develop a thin skin when exposed to air, for 3D printing. The skin thus formed is said to be strong enough to hold the liquid's shape.

"It's difficult to create structures out of liquids, because liquids want to bead up. But we've found that a liquid metal

alloy of gallium and indium reacts to the oxygen in the air at room temperature to form a 'skin' that allows the liquid metal structures to retain their shapes," says Dr Michael Dickey, an assistant professor of chemical and biomolecular engineering at NC State and co-author of a paper describing the work.

The researchers worked on multiple techniques for creating these structures, which can be used to connect electronic components in three dimensions. While it is relatively straightforward to pattern the metal on the same level, these liquid metal structures can also form shapes that reach up or down.

Long-lasting batteries built from rice husk

Lithium-ion (Li) batteries play an important role in today's technological progress as they power the electronics and hybrid electric vehicles industries. With an aim to increase battery capacity, many researchers have so far struggled to replace the carbon anode in today's technology with a silicon anode which has the capacity to hold ten times more charge. Next-generation high-capacity lithium-ion batteries thus developed are expected to speed the development of more sophisticated portable electronics.

A team of scientists from South Korea has now proposed to use layers of silicon dioxide (silica) as a source of silicon. Usually, the silica-rich husks left over after rice is harvested are turned into fertiliser additives. These silica layers are said to be uniquely porous at the nano-scale level.

The team chemically converted the rice husk silica into pure silicon and then fashioned battery electrodes out of the material. "It showed no capacity fade even after 200 charge-discharge cycles. A synthetic silicon electrode used for comparison had a higher initial charge capacity but faded badly; it began performing worse than the rice-husk electrode after 10 to 15 cycles," the report added.

Solar car designed by students

A group of students from Manipal Institute of Technology has designed a commercial solar-powered car, SolarMobile, which has been priced at ₹ 1,600,000. Right now a plastic chassis is being tested for the car which is supposed to hit the street by February 2014.

"This is a part of our project work and is supported by the Ministry of Renewable Energy and some private companies. We got ₹ 700,000 from the government and ₹ 1,300,000 from other sponsors," the team was quoted as saying by *The Times of India*.

"It had gone up to 85 kmph during tests on campus. We plan to take part in the South Africa Solar Challenge in September next year," said Hrishikesh Rao, a second-year student of the mechatronics department.

"The car will cost around ₹ 800,000 if produced on a mass scale," said Anjan Kumar, a third-year student of instrumentation and control engineering. "Our first car, Arka, was solar powered too, but it had a single seat and was not so good looking."